

Review of summation notation: Let $x_1, \dots, x_n$ denote a sequence of numbers; $y_1, \dots, y_n$ denote another sequence of numbers; and a denote a constant. Then, we have

$$\sum_{i=1}^n ax_i = a \sum_{i=1}^n x_i, \quad (1)$$

$$\sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i, \quad (2)$$

$$\sum_{i=1}^n a = n \times a. \quad (3)$$

Answer the following questions:

1. Explain the idea of OLS estimation and the related minimization problem.
2. Derive the formula of the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ following the steps in the lecture note.
3. Show the following holds:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}, \quad (4)$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. (Hint: show $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$ and $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$.)

4. Suppose that a researcher, using data on class size (CS) and average test scores from 50 third-grade classes, estimate the OLS regression:

$$\widehat{TestScore} = 640.3 - 4.93 \times CS. \quad (5)$$

- (a) A classroom has 25 students. What is the OLS regression's prediction for that classroom's average test score?
- (b) Last year a classroom had 21 students, and this year it has 24 students. What is the OLS regression's prediction for the change in the classroom average test score?
- (c) The sample average class size across the 50 classroom is 22.8. What is the sample average of the test scores across the 50 classrooms? (Hint: Review the formulas of the OLS estimators.)